

Chapter 17

Covariation Between LULC Change and Hydrological Balance in River Basin Scale



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Abstract Land use and land cover (LULC) change has one of the key modes of human modifications and has raised several queries related to its impact on hydrology and climate. Since the LULC plays a vital role in partitioning energy and water fluxes at the land surface into different components, the changes in LULC can impact the water and energy cycles to a significant level. However, at a basin scale, the severity of such consequences depends on the scale, type, and heterogeneity of LULC changes. An assessment of the impacts of LULC change on hydrology in three major river basins—Brahmaputra, Ganga, and Mahanadi of India is performed in this study using the variable infiltration capacity (VIC) model, a physically-based distributed hydrological model. The assessment reveals an important compensating effect in the hydrologic changes resulted from LULC transformations. The negative consequences (e.g., increased surface runoff due to urbanization) at one place are compensated by opposite positive changes (e.g., decreased surface runoff due to forest plantation) at other places at the basin scale. Such compensation leads to insignificant hydrologic changes at the basin scale; however, the consequences are significant at the local and sub-basin scales.

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17.1 Introduction

The land use land cover (LULC) is a boundary layer between the earth's subsurface and the atmosphere, which governs the partitioning of energy and water fluxes at the land surface. With the increased demand for land and urban infrastructures to cater increasing population, the natural LULC has been transformed into managed lands such as agriculture and builtup (Li et al. 2015; Patidar and Keshari 2020; Tsarouchi et al. 2014). Such modifications have led to significant physiological variations due to altered vegetation type, coverage, and density (Jacobson 2011; John et al. 2016). Conversion of LULC types results in altered canopy coverage, rooting depth, albedo, surface roughness, and biomass. Moreover, the LULC changes modify soil compactness, porosity, infiltration capacity, hydraulic conductivity, and organic contents of the topsoil layer (Ghimire et al. 2014; Liu et al. 2002). Consequently, the partitioning of incoming solar radiation into sensible and latent heat fluxes, precipitation to runoff and infiltration, wind speed, soil moisture, and heat storage capacity is changed (Mishra et al. 2010; Patidar and Behera 2019). These direct and indirect impacts of LULC change can negatively affect the hydrology and climate at local and regional scales (Bosmans et al. 2017; Sterling et al. 2013). A few severe alterations in hydrology and climate induced by LULC changes observed at different parts of the world include (1) increased surface runoff and flood frequency due to deforestation and urbanization (Schilling et al. 2010; Yang et al. 2010), (2) increased evapotranspiration (ET) due to agriculture expansion (Jaksa and Sridhar 2015; Lei et al. 2015), (3) changes in baseflow contribution to rivers due to deforestation (Das et al. 2017; Peña-Arancibia et al. 2019), (4) decreased groundwater recharge due to increased impervious surfaces associated with urbanization (Minnig et al. 2018), and (5) decreased diurnal temperature due to urbanization (Kalnay and Cai 2003). Although, there are mixed opinions on the consequences of LULC changes such as on the baseflow, temperature, and precipitation.

Since the LULC plays a vital role in exchanging water between the surface and the subsurface through infiltration and root water uptake, the hydrologic impacts' assessment became important during the past few decades. Rapid urbanization and agriculture expansions made the LULC change an essential consideration in water resources assessment and allocation, flood and drought forecasting, future planning for sustainable water management, etc. LULC changes can affect water availability by altering streamflow and groundwater (Dawes et al. 2012; Schilling et al. 2010). The impervious surfaces can increase the surface runoff and decrease groundwater recharge by blocking the infiltration, leading to short-term increased streamflow during the rainy season, and decreased streamflow and groundwater during the dry season. Such changes also lead to increased flood magnitude and frequency (Bloschl et al. 2007). Similarly, deforestation results in increased surface runoff due to decreased canopy interception and ET. The consequences of LULC changes are serious when the hydrologic responses of the transformed LULC are considerably different from that of the previous LULC distribution. Generally, the conversion of forest to settlement is the most sensitive to affect hydrology; although, the magnitude

of such impacts varies due to other factors such as topography and climate (Patidar and Behera 2019). The precipitation characteristics may alter the sensitivity of the LULC changes to its effects on hydrology, where the impacts are more prominent in the areas with uniform temporal distribution (Wagner et al. 2013). It implies that hydrologic changes could be different even with similar LULC transformations in different basins.

The Indian River basins mostly receive precipitation through the southwest monsoon. The spatio-temporal distribution of precipitation in India is highly erratic, which induces variable hydrologic responses in the different river basins. Moreover, due to high heterogeneity, the multidirectional changes are evident in large Indian River basins. Therefore, assessing the sensitivity and impacts of LULC change to hydrology under Indian climatic conditions observed during the past few decades is necessary for designing effective policies for sustainable water management. The key objectives of this study were to investigate the sensitivity of various LULC transformations and assess their impacts on the surface runoff and ET in three major Indian River basins as Ganga, Brahmaputra and Mahanadi, and east-flowing river basins.

17.2 Study Area and Data Used

The study area includes three major river basins—Ganga, Brahmaputra, and Mahanadi—and other associated eastern flowing river basins such as Brahmani-Baitarani, Subarnarekha, Nagavali, and Vamsadhara of India (Fig. 17.1). The assessment was done only for the Indian region of the basins having a total geographic area of 10, 26, 938 km², covering approximately 31% of the total geographic area of India. The Ganga–Brahmaputra basin extends from Himalaya in the north to the Bay of Bengal in the south covering diverse topography, LULC, and climate. The elevation ranges of the study area vary between 0 and ~7322 m (above m.s.l). The climate condition varies from per-humid in the southeast to arid in the east (Raju et al. 2013). The climate shows significant variations in space and time, with a daily average temperature varying between −5 °C and 27 °C. The precipitation mostly occurs during mid-June to September, and the total annual ranges from 543 mm (in the west) to more than 2300 mm (in the northeast). The natural vegetation varies with latitude from deciduous to evergreen forests.

The Mahanadi and east-flowing river basins, hereafter referred to as MEB, are bounded by Ganga in the north, the Bay of Bengal in the east with a complex drainage system near the coast. The average annual precipitation in MEB is ~1360 mm, with an average temperature ranging from 4 °C (winter) to 42 °C (summer). The monsoonal precipitation accounts for ~86% of total annual precipitation in MEB.

Various remote sensing and in-situ datasets were used in this study. The gridded precipitation and temperature data generated by the India Metrological Department (IMD) were used. These datasets are daily gridded precipitation (0.25°) and temperature (1°) derived from an extensive network of observation sites (Pai et al. 2014). The altitude and related information, such as slope and river networks, were extracted

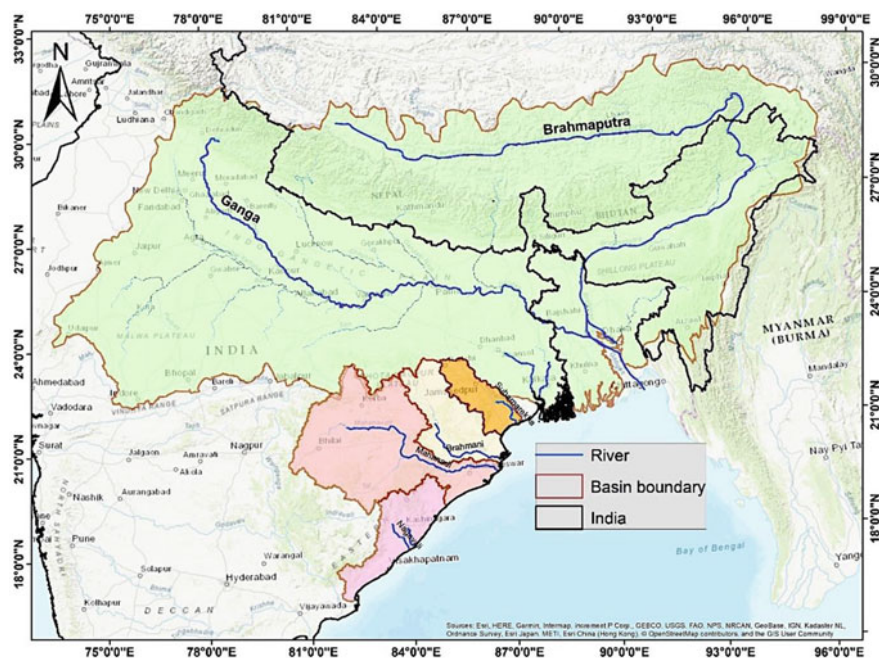


Fig. 17.1 Ganga, Brahmaputra, Mahanadi, and basins of east flowing rivers including Brahmani-baitarani, Subarnarekha, Nagavali, and Vamsadhara. The study area is the Indian part of these river basins

from the shuttle radar topographic mission (SRTM) digital elevation model (DEM). The soil maps of the Food and Agriculture Organization (FAO) and the National Bureau of Soil Survey and Land Use Planning (NBSS and LUP), India were used. The vegetation parameters, such as leaf area index (LAI) and albedo, were extracted using moderate resolution imaging spectroradiometer (MODIS) LAI and albedo data product. The stream discharge data of the Central Water Commission (CWC) were used for model calibration and validation in the Mahanadi River basin.

The LULC maps derived by Behera et al. (2014) for Ganga, Roy et al. (2015) for the Brahmaputra, and Das et al. (2020) for MEB were used. The LULC maps for Ganga were available for the years 1975 and 2010, and for Brahmaputra and MEB for the years 1985 and 2005.

17.3 Methodology

The hydrologic impacts of LULC change can be assessed either using long-term in-situ observations or hydrological modeling. Although the former method provides a more realistic scenario, it is rarely used due to the unavailability of long-term

observations and the influence of other factors like climate variability. The observed changes in hydrologic fluxes need to be precise to get convincing information about the changes (Ashagrie et al. 2006). On the other hand, hydrological modeling provides a systematic way of estimating the impacts of LULC changes. This method is commonly used to simulate the differential effects of climate conditions and LULC changes on hydrologic parameters; wherein, the accuracy is subjected to the precision of inputs, parameters, and efficacy of the model.

Here, the variable infiltration capacity (VIC) hydrological model is used. Simulations were performed taking the LULC maps of two time periods keeping the other variables and parameters unchanged. The LULC map of 1975 for Ganga and 1985 for Brahmaputra and MEB was used in the first simulation, while the LULC map of 2010 for Ganga and 2005 for Brahmaputra and MEB was used in the second simulation. The simulated ET and runoff were analyzed for quantitative and comparative assessment. This method assures that the change in ET and runoff is isolated from human interventions as water diversions, storages, and climate variability. The following sub-sections elaborate on the working principle of the VIC model and its implementation in the study area.

17.3.1 VIC Model Description

The VIC model is developed by the University of Washington and is based on the concept of variable infiltration curve developed by Ren-jun et al. (1980). The model has evolved substantially since its development in 1990's and has been tested in various river basins throughout the world (Gao et al. 2009; Ren-jun et al. 1980). VIC is a grid-based model in which the simulations for each grid are performed independently, and a routing model is coupled externally to derive the streamflow (Lohmann et al. 1998b). VIC is capable of handling the sub-grid heterogeneity by considering the proportions of LULC within the grid cells, which make it suitable for assessing the impacts of LULC change. This model also includes multiple soil layers for a better simulation of soil moisture and water infiltration. The other inputs, such as meteorological and topographic variables, are also provided at a grid scale. The ET is simulated using the Penman–Monteith equation. Thus, the ET is sensitive to any change in vegetation parameters such as LAI, albedo, root depth, and stomatal resistance. The parameters for respective LULCs within the grids are read from the look-up table. ET over a grid cell is simulated as the sum of the evaporation from the canopy layer of the n th vegetation tile, transpiration from the n th vegetation tile, and evaporation from the bare soil weighted by the respective surface cover fractions. The canopy evaporation from the n th vegetation tile is estimated as a function of potential ET and other resistances weighted by the vegetation fraction. A few additional factors on canopy resistance and soil moisture are added to estimate transpiration from the respective vegetation tile. While the runoff is estimated using the ARNO model (Todini 1996).

17.3.2 Assessment of the Hydrologic Impacts Using VIC Model

The model was run to simulate ET and runoff under two LULC scenarios, and change was detected by subtracting the fluxes. The overall schematic of the methodology is shown in Fig. 17.2. A two-layer model was developed, wherein the soil parameters were defined using the soil texture information from NBSS and LUP (for the top layer) and FAO (for the bottom layer) soil texture maps. The soil hydraulic properties, such as saturated hydraulic conductivity, field capacity, and wilting point, for each soil texture were defined using the index provided by Cosby et al. (1984). Vegetation parameters are based on the Global Land Data Assimilation System (GLDAS) as described by Rodell et al. (2004). The LAI value, which is the most sensitive vegetation parameter to alter water balance in the VIC, is extracted from the moderate resolution imaging spectroradiometer (MODIS) LAI data product [MCD15A3] (Patidar and Behera 2019). The vegetation parameters are provided on a monthly scale irrespective of the soil type. The changes in LULC are addressed by changing the LULC fractions under each scenario while keeping the same parameter table (Fig. 17.2).

The simulations in each grid are performed independently in VIC and induce no flux exchange from/to the neighboring grids. The grids are interconnected only during the post-processing, where the VIC-simulated runoff is estimated at the basin outlet using the routing module (Lohmann et al. 1998a). The simulations can be done without considering the entire basin, although the entire catchment is required when estimating streamflow. Therefore, the VIC was suitable for this study as only

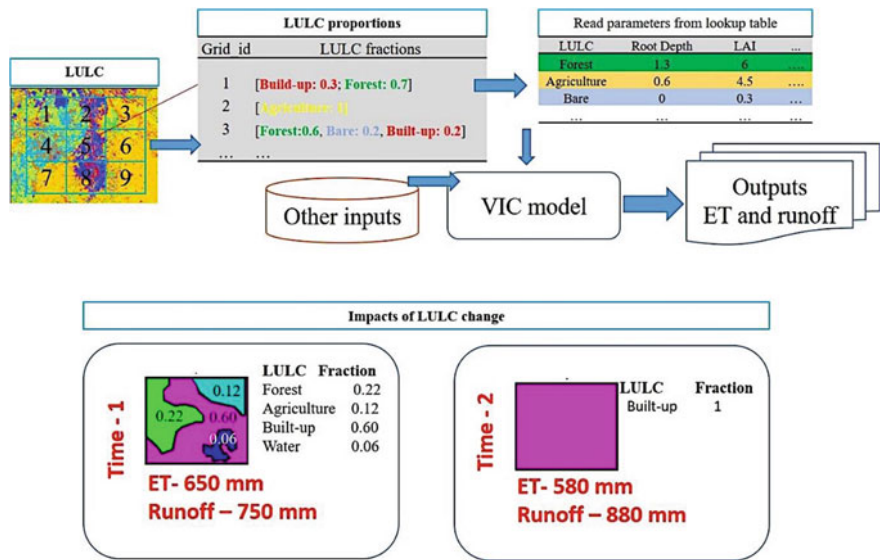


Fig. 17.2 Schematic of methodology for assessing hydrologic impacts of LULC change

the Indian part of Ganga and Brahmaputra River basins were considered due to data constraints. The model was calibrated and validated in the Mahanadi River basin using the streamflow data accessed from the Central Water Commission (CWC) at Basantpur, Sundergarh (from 2001 to 2010), and Rhondia (from 1971 to 1979) gauging sites. The Nash–Sutcliffe efficiency (NSE) was estimated as 0.56 and 0.65 at Basantpur and Sundergarh, respectively, during calibration. During the validation, the NSE is observed to be 0.76 at Rhondia, which shows an acceptable performance.

17.4 LULC Transformations

Agriculture is the most dominated LULC in all the river basins, except the Brahmaputra, dominated by forest covers. Agriculture occupies 73% of Ganga, 55% of MEB (Mahanadi and eastern flowing river basins), and 16% of Brahmaputra River basin, as per the recent assessment done by Behera et al. (2014, 2018). The Brahmaputra basin is dominated by forest, which occupies 62% of the total geographic area of the basin. The Ganga and MEB also accommodate considerable forest areas. Ganga has 20% and MEB has 35% of the area under forest. A generalized LULC map is provided in Fig. 17.3. A detailed discussion on the LULC of the study area can be found in Behera et al. (2014, 2018).

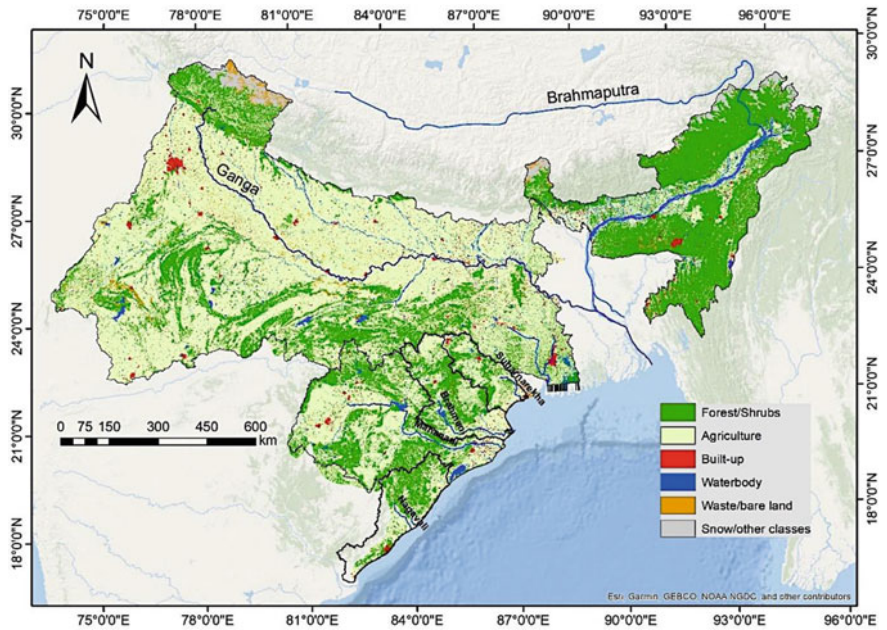


Fig. 17.3 LULC of the study area (2010 for Ganga) and 2005 for other river basins

The key transformations during the past, from 1975 to 2010 in Ganga and 1985 to 2005 in Brahmaputra and MEB, include deforestation and the expansion of agricultural and urban areas. Figure 17.4 highlights the location and extent of these changes. Although the buildup has a small coverage with respect to the total geographic area of the basins, it shows a considerable percentage increase in almost all basins. In Ganga, an increase of 45% built-up area is observed while the MEB and Brahmaputra show an increase of 16% and 6%, respectively. Deforestation is another important change which occurred in all the basins. A net change of 4% in the forest in Ganga, 1.5% in MEB, and 1.6% in the Brahmaputra is observed. A common trend has been to expand the agricultural lands at the cost of the nearby wastelands and forest/shrublands. The agriculture and waste lands also lost a considerable proportion to built-up lands. All changes are multidirectional at basin scale, except the built-up expansion. There has been a decrease in green areas due to deforestation in one place, while an increase at other places due to forest regeneration, afforestation, and plantations is observed. Such complementary changes lead to a small net change when analyzed at the basin scale. Therefore, the changes are more visible at a local scale, for instance, there are many patches of deforestation as shown in Fig. 17.4; however, the net change in the forest is less due to afforestation and plantations at other places.

The change map also indicates that deforestation is the most common change in all the basins. The other changes are relatively more evident in the Ganga basin as compared to other basins. These changes are mostly driven by the need for land for

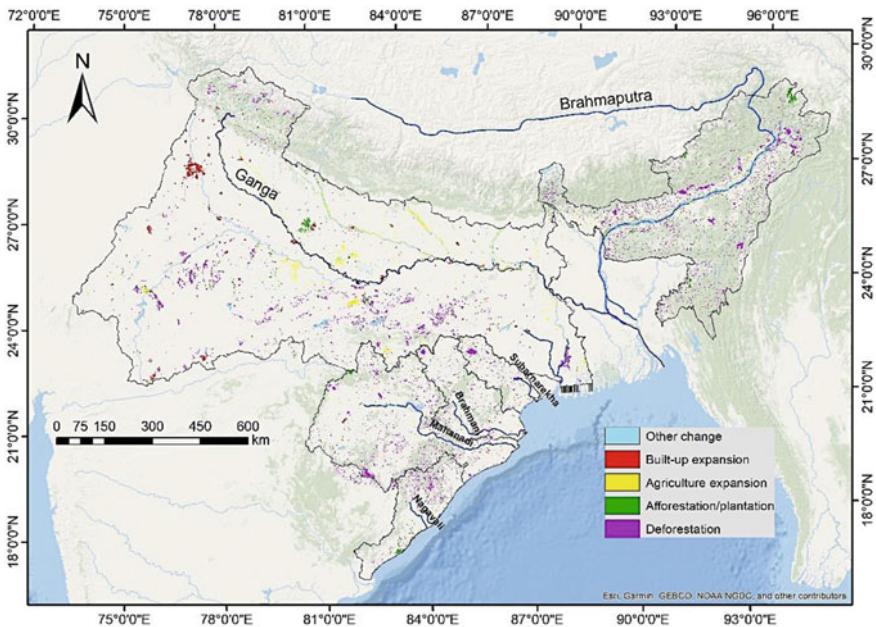


Fig. 17.4 Major LULC transformations during the past decades

agriculture and urban infrastructures. The Ganga has the most fertile lands among all basins with suitable topography, climate and availability of surface water, and groundwater resources for irrigation. The suitability of land for agriculture could be an important driving factor behind the expansions of agriculture patches at the cost of nearby natural vegetation or waste lands. An analysis was done by Behera et al. (2014) to investigate the change in agriculture patches of different sizes between 1975 and 2010, and it was found that the mean patch size of agriculture has been increased from 0.76 to 0.82 for small patches ($<50 \text{ km}^2$). Similarly, the urban expansion has also been the highest in the Ganga basin than the other two basins. Being the most populated basin, accommodating 40% of India's population, the Ganga basin has the highest human settlements. Expansions at the periphery of such existing settlements with an increased density led to a significant increase in built-up areas (Patidar and Keshari 2020).

17.5 Potential Impacts

Quantification of the potential impacts of LULC change is important for making effective policies for sustainable water management. An experiment was performed using the VIC model to investigate the effect of various LULC transitions on ET and runoff under Indian climatic conditions. The simulations are performed by changing the LULC in the model grid literally for 10 LULC classes using constant climatic conditions at $23^{\circ}15'1'' \text{ N}$ and $83^{\circ}14'59'' \text{ E}$. Using the daily simulations from 1975 to 2005, the annual average ET and runoff were calculated and compared for each LULC class.

The change matrix of the impacts is shown in Fig. 17.5. The conversion from vegetation to non-vegetation shows the highest sensitivity to affect the ET and runoff. A conversion from evergreen broad-leaved forest (EBF) to builtup (BU) can reduce the annual ET up to 377 mm, which is the highest among all conversions. The conversion from other forest classes, such as evergreen needle-leaved (ENF), mangroves

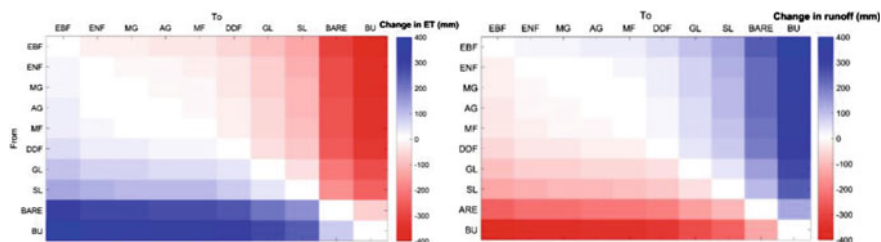


Fig. 17.5 Change matrix showing the hydrologic responses of various LULC transitions. EBF: evergreen broad-leaved forest, ENF: evergreen needle-leaved forest, MG: mangroves, AG: agriculture, MF: mixed forest, DDF: deciduous forest, GL: grassland, SL: scrubland, BARE: bare ground, BU: builtup

(MG), mixed forest (MG), and deciduous forest (DDF), to bare/built up also leads to a significant change in ET as highlighted by red color in the matrix (Fig. 17.5). Conversion from one vegetation class to the other vegetation class is relatively less sensitive. The impact on runoff is just opposite to that on the ET. The highest impact is observed due to evergreen broad-leaved forest (EBF) conversion to builtup (BU), which can increase the annual runoff up to 377 mm. Such alterations could impact the streamflow and flood magnitude to a considerable extent. Similar to ET, the transitions from vegetation to non-vegetation, or vice-versa, have the largest impact on runoff.

Theoretically, the magnitude of impacts depends on multiple factors including soil hydraulic properties, meteorological conditions, and topography. In this experiment, the other factors were kept unchanged to understand the LULC transition-wise impacts. However, it is obvious to have slight variations in the magnitude of the impacts when some of the aforementioned factors are changed; although, the order of sensitivity of impacts is expected to be the same. It is also known that precipitation is the most sensitive to affect runoff (Bloschl et al. 2007). Similarly, the soil moisture availability in the root zone is among the key governing factors for ET (Jaksa and Sridhar 2015). Therefore, under real conditions, when the influence of all these factors is combined, the impacts of small LULC change might not be very significant.

17.6 Changes in ET and Runoff

The ET and runoff simulated using the LULC maps of two time periods were compared to study the effect of LULC change (Figs. 17.6 and 17.7). In the Ganga basin, annual ET has a maximum increase of 74 mm and a maximum decrease of 123 mm. An opposite impact was observed on the runoff with a maximum increase of 122 mm to a decrease of 73 mm. In the Brahmaputra basin, the range of change in ET was observed between -99 and 7 mm (where negative indicates a decrease). The MEB has the lowest impact on ET, which varies between -4 to 5 mm. An opposite impact on runoff is observed. The runoff change varies between -73 to 122 mm in Ganga, -21 to 13 mm in the Brahmaputra, and between ± 2 mm in MEB. Although the impacts are evidently visible at a grid scale, the impacts are minimal at the basin scale. The net change in ET and runoff in Ganga is found to be insignificant (approximately 1 mm). Similarly, the net impacts are within 1.5 mm in the Brahmaputra and within 0.5 mm in MEB.

It is seen in Brahmaputra and MEB that the impacts on ET are higher as compared to runoff. However, in Ganga, the magnitude of the changes in ET and runoff is comparable. This indicates the role of precipitation characteristics in the impact assessment of LULC change. The differential impacts in the Ganga and Brahmaputra/MEB could be attributed to a higher number of precipitation events in Brahmaputra and MEB, which needs to be studied further. The more number of rainy days

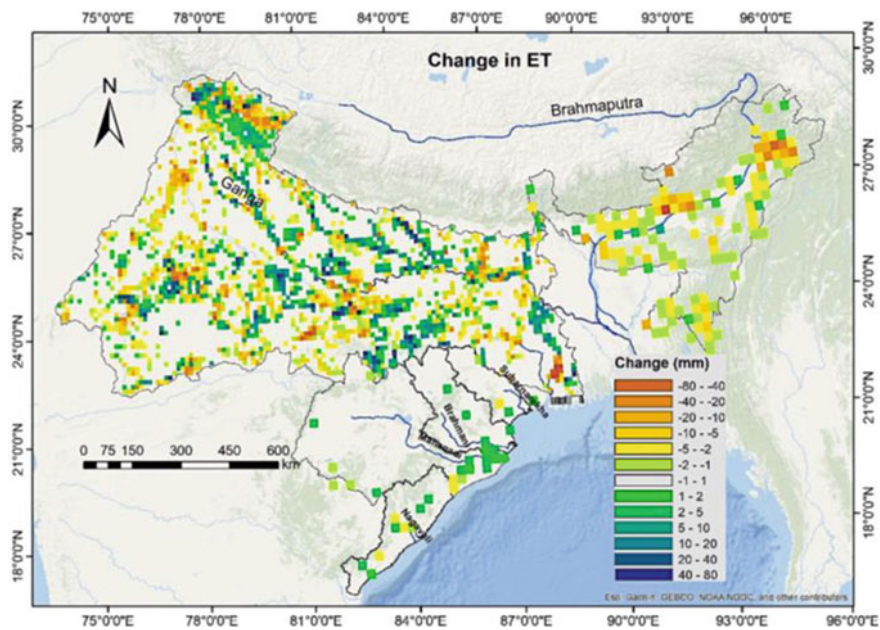


Fig. 17.6 Changes in ET induced by LULC change

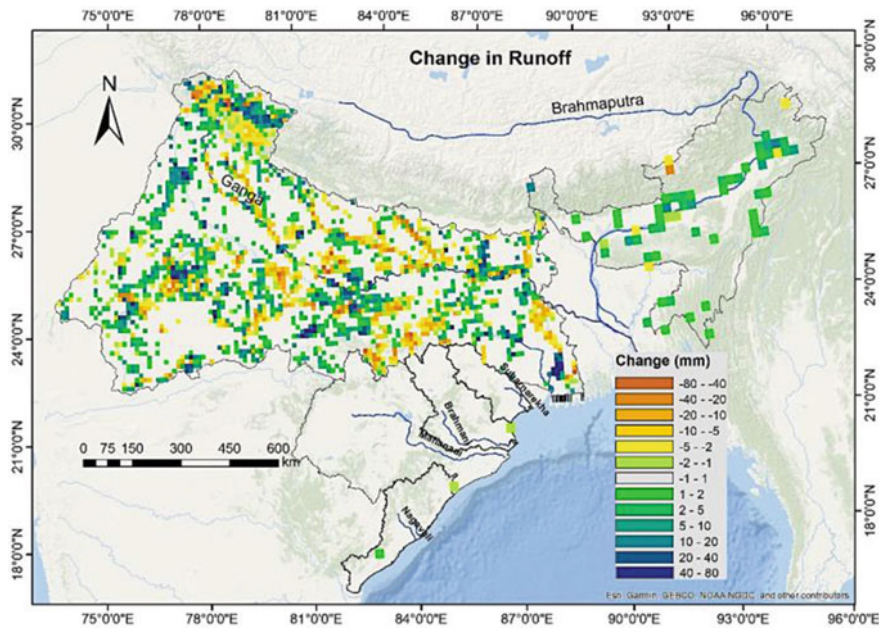


Fig. 17.7 Change in runoff induced by LULC change

in a year can provide more soil moisture to plants leading to a higher ET from the same vegetation.

17.7 The Compensation Effect

The assessment of the impacts in Indian River basins showed insignificant impacts on ET and runoff despite various changes in LULC, which highlighted a very important mechanism of compensating negative impacts by positive impacts. Due to multidirectional LULC transformations in all the basins, the resulting increase in ET/runoff at one place was nullified by the decrease in ET/runoff at other places within the basin. The decrease in ET induced by deforestation was mainly compensated by increased ET resulted from plantations. The conversion from waste/bare land led to increase ET, which also reduced the overall impact of deforestation. The histograms of ET and runoff are plotted to understand this mechanism (Fig. 17.8), showing the change in ET/runoff on abscissas and the number of grids on the ordinate. The magnitude and the extent of changes are clearly highlighted in these plots. The bars in the left indicate a decrease (negative) and in the right indicates an increase. A perfect symmetry of the histogram indicates all increase is compensated by the decrease, which is true for Ganga and MEB. On the other hand, the Brahmaputra shows slight skewness toward negative for ET and toward positive for runoff, indicating a net decrease in ET and increase in runoff. It can also be observed that most of the changes are of small magnitude while the large magnitudes have small frequencies in all the basins,

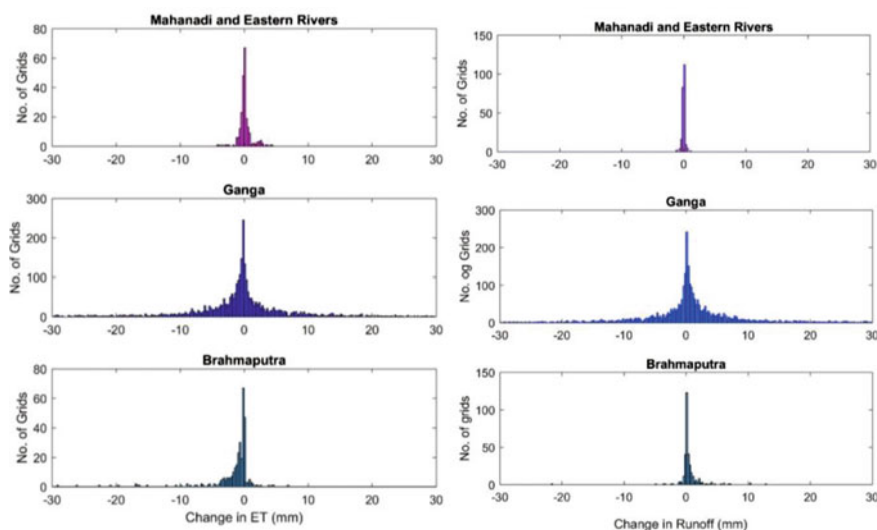


Fig. 17.8 Histograms of change in ET and runoff

as indicated by the near-zero peaks. This is another reason behind the small net impacts.

In all the basins, the decrease in the ET induced by the conversion of forest/agriculture to wasteland/builtup/scrubland was mostly compensated by the increase in ET due to increased plantation/agriculture areas at the basin scale. However, the impacts are more prominent at a local scale, for instance, at the sub-basin level, the changes in annual ET range from a decrease of 8.5 mm to an increase of 9.2 mm, which shows that the small sub-basins are more sensitive to LULC changes due to less compensation effect (Patidar and Behera 2019). A similar phenomenon was observed in Meuse River basin in Western Europe (Ashagrie et al. 2006). Similarly, an assessment of the impacts of LULC change in an agriculture catchment, named Mula and Mutha, upstream of Pune, India, using soil and water assessment tool (SWAT), also revealed that the impacts are too small at the basin level due to compensation effects, even after substantial LULC changes (Wagner et al. 2013).

17.8 Conclusions

The knowledge about the LULC induced changes in the hydrological regime has become important in the changing world due to increasing concern about water availability and hydrological disasters. In this context, the information about the shifts in the hydrological regime of major Indian River basins—Ganga—Brahmaputra and Mahanadi—could be an important input to the upcoming water management policies.

The assessment of the LULC induced changes in ET and runoff in these river basins reveals a very important compensating effect occurring due to multidirectional LULC changes. The negative consequences are nullified or at least reduced by the positive changes in ET and runoff at the basin scale. Fortunately, despite various LULC alterations, this compensating effect led to insignificant changes at basin scale in almost all river basins. The net change in ET and runoff is within 1.5 mm/year. However, the changes are significant at local scales and range up to 123 mm/year. This indicates that a bottom-up approach would be more effective for managing natural landscapes and dependent systems. This would require more detailed investigations of the changes in LULC and hydrology at watershed and sub-basin levels.

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